

National University of Computer & Emerging Sciences, Karachi

Department of Computer Science

Fall-2025

CS2009: Design and Analysis of Algorithms

Assignment 5

Due Date: 7th December 2025

20% penalty for 1 day late

40% penalty for 2 days late

Submission is not allowed afterward

Max Marks: 100 Marks

Question#01: Watch the video lecture on [P, NP, NP-hard, and NP-complete problems](#). And write the answer to the following question in your own words. **[70 Marks]**

- What is meant by P and NP Problems? Explain $P = NP$.
- Why it is important to find approximate solutions for NP-Complete Problems.
- What is the difference between NP-Hard and weakly NP-hard class problems?
- What is the 3-SAT problem?
- What is meant by NP-complete problems? How can we prove a problem is NP-complete? Explain with an example.
- What is Reduction?
- A problem that is solvable in time complexity of $T(n)=3*n^n$ and space complexity of $S(n)=n^2$ and it can be validated in $T(n)=2^n$ time. Is it an NP-Complete or NP-Hard? Explain.

Question#02: Consider the following APPROX-VERTEX-COVER algorithm. Proof that this algorithm is a 2approximation method for VERTEX-COVER. **[15 Marks]**

```
APPROX-VERTEX-COVER(G)
C = ∅;
E' = G.E;
while(E' ≠ ∅){
  Randomly choose a edge (u,v) in E', put u and v into C; Remove all the edges
  that covered by u or v from E'
}
Return C;
```

Question#03: An Instance (X, F) of the set-covering problem consists of a finite set X and a family F of the subset of X, such that every element of X belongs to at least one subset of F:

$$X = \bigcup_{S \in F} S$$

We say that a subset $S \in F$ covers all elements in X. Our goal is to find a minimum size subset $C \subseteq F$ whose members cover all of X.

$$X = \bigcup_{S \in C} S$$

Algorithm 1: GREEDY-SET-COVER (X, F)

```
1  $U \leftarrow X$ 
2  $C \leftarrow \emptyset$ 
3 While  $U \neq \emptyset$ 
4   do select an  $S \in F$  that maximizes  $|S \cap U|$ 
5      $U \leftarrow U - S$ 
6      $C \leftarrow C \cup \{S\}$ 
7 return  $C$ 
```

Consider each of the following words as a set of letters: {arid, dash, drain, heard, lost, nose, shun, slate, snare, thread}. Show which set cover GREEDY-SET-COVER produces when we break ties in favor of the word that appears first in the dictionary. **[15 Marks]**

****Best of luck****